

Which surface should stop the sliding block sooner?

Learning intention

Compare a sliding block's movement on different surfaces and explain the effect of friction using a controlled test.

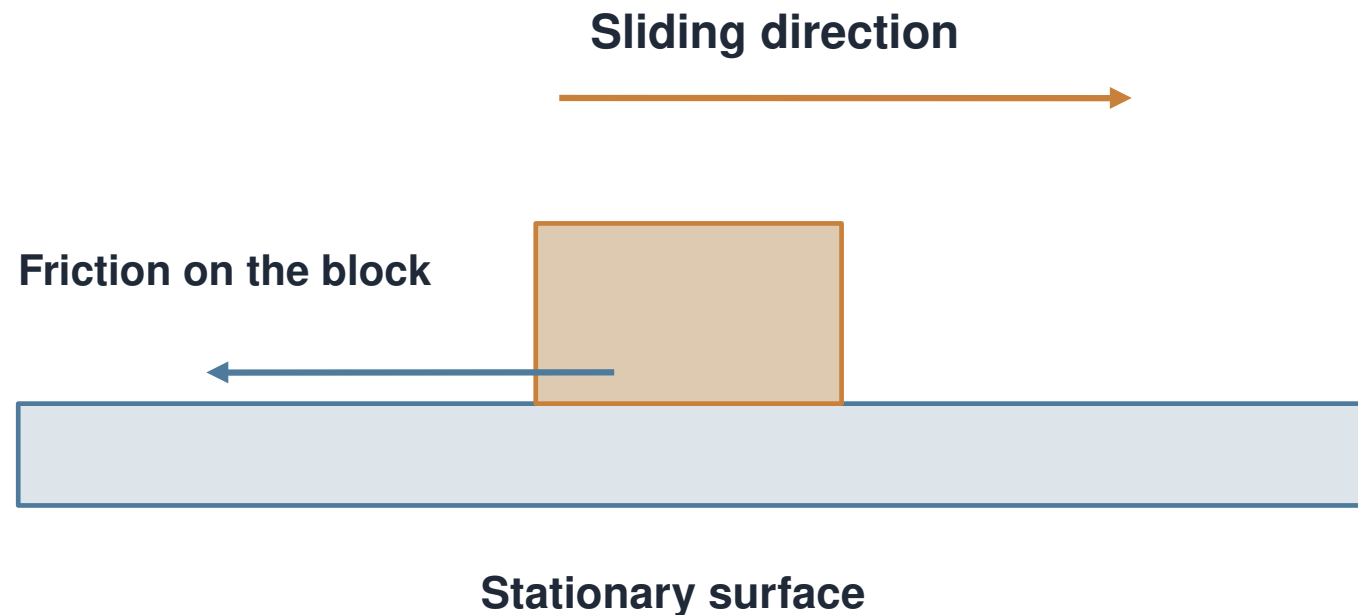
Predict from evidence.

Keep the release the same.

Retrieve pushes, pulls and comparisons

1. What can a push change?
2. Name one thing to keep the same in a test.
3. What unit can measure a short distance?

I distinguish motion from force

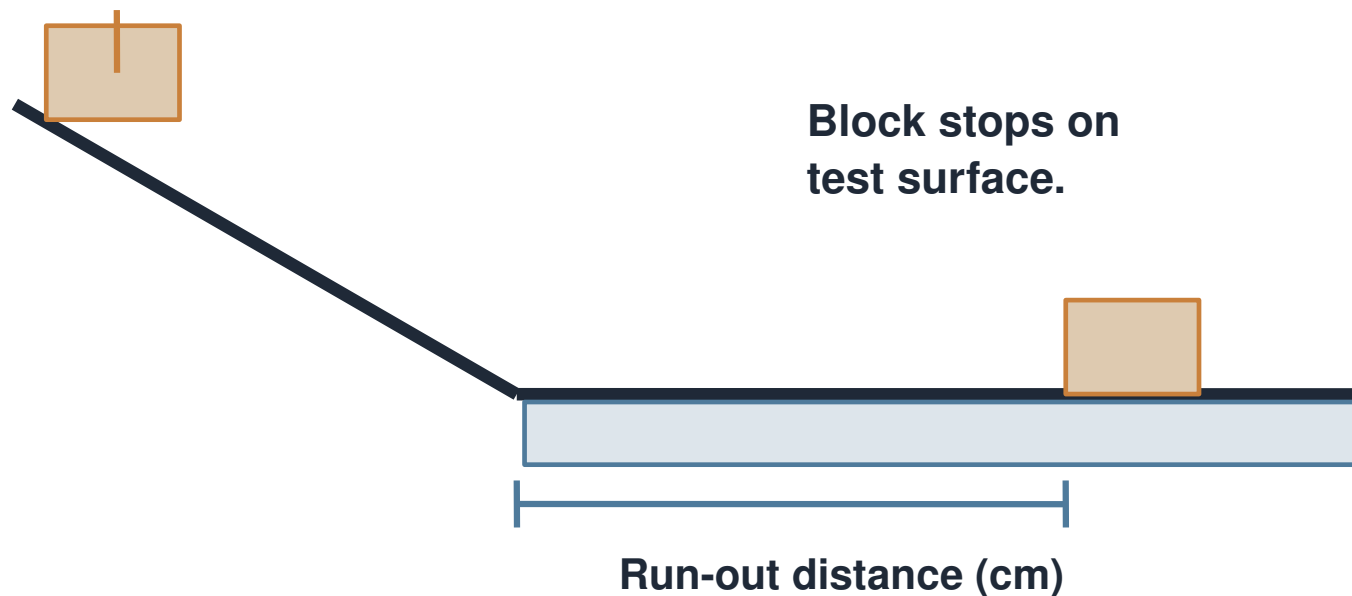


After release: no hand keeps pushing the block.

The block slides right. Friction on it acts left. Contact with the surface matters.

I keep the start conditions fixed

Same ramp and release mark

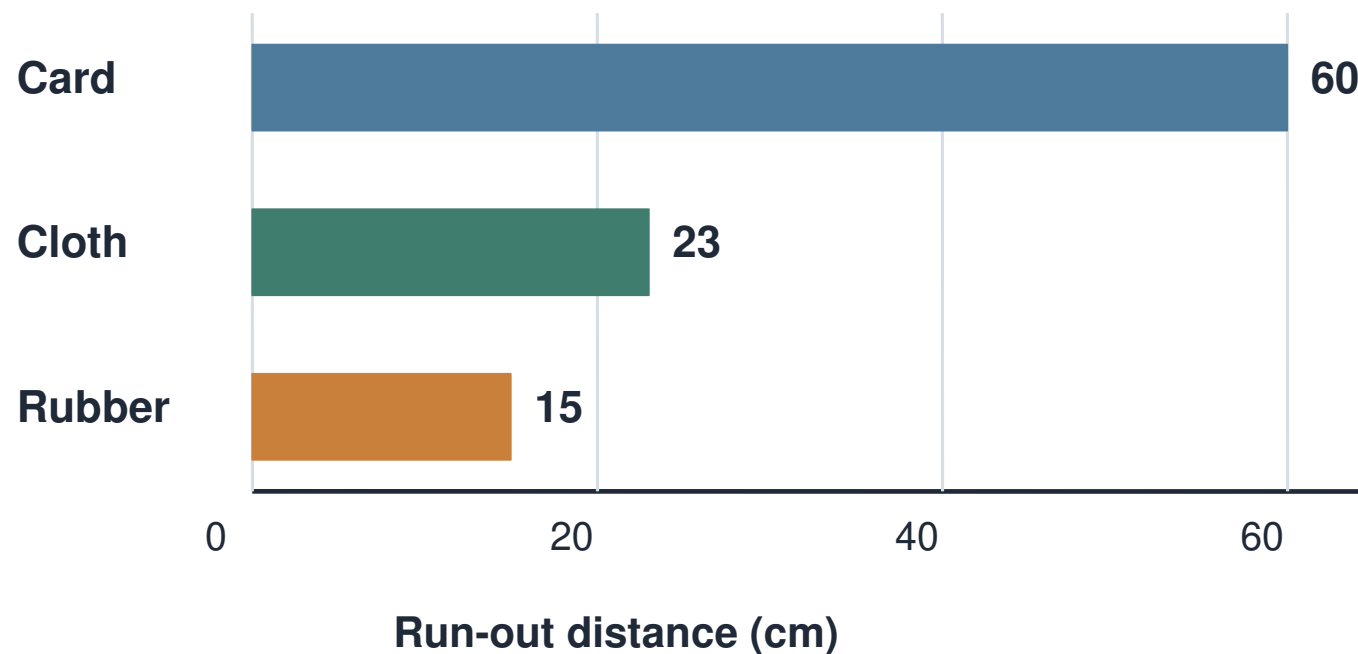


Use the same trailing edge at ramp end and when stopped.

Same block, face and ramp. Same release mark; no push. Change only the level surface.

I use repeats before choosing

Constructed mean run-out distances



Compare all three runs. A longer run-out means less slowing here.

The surface-selection meeting

Read the three runs. Choose the shorter stopping zone. Use two readings in your reason.

Surface	Run 1 (cm)	Run 2 (cm)	Run 3 (cm)
Card	58	60	62
Cloth	22	23	24
Rubber	14	15	16

Constructed sliding-block data; same ramp, block face and release

Repair a changed release

A team pushes only the rubber run.

Can we compare it fairly?

Name the repair.

The block slides right after release.

Which friction statement fits this setup?

- A** Friction on the block acts left, opposing sliding.
- B** Friction acts right because the block moves right.
- C** No force can affect the block after the hand lets go.

Choose your science route

Read and select a surface.

Write an evidence-based recommendation.

Review the strength of the test.

Make your own evidence record

Use the shared evidence.

Explain one choice.

Check a label or number.

Check what the evidence can show

Point to the clue you used.

Remove an unsupported claim.

Lundy loop: improve the shared account

Choose how to share.

Hear one evidence-based challenge.

Improve or defend one idea.

What made our surface choice defensible?

Keep the useful change

What changed in our account?

Which evidence helped?

Three final checks

E1. What does friction oppose in this test?

E2. Which supplied surface gives the shortest run-out?

E3. Name one start condition to keep the same.